

Alok Raj

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EDUCATION

VIT Bhopal University

B.Tech in Computer Science and Engineering (Cloud Computing and Automation)

Sep 2023 – May 2027

CGPA: 8.93/10

G.D. DAV Public School, Deoghar

Class XII (CBSE)

Jul 2020 – Jun 2022

Percentage: 89.2%

St. Francis School, Deoghar

Class X (ICSE)

Mar 2008 – Jun 2020

Percentage: 85.6%

TECHNICAL SKILLS

Core CS: Data Structures & Algorithms, Object-Oriented Programming, Operating Systems, DBMS, Computer Networks

Languages: Java, Python, JavaScript, SQL

Systems & Backend: Distributed Systems, Microservices, Service-Oriented Architecture, REST APIs

Cloud & Infrastructure: AWS (Lambda, Step Functions, API Gateway, DynamoDB, S3, EC2, CloudWatch, EventBridge, IAM, Bedrock, Rekognition, CloudFormation)

Databases: MongoDB, MySQL, DynamoDB

Frontend: React.js, HTML5, CSS3

DevOps & Tools: Docker, Kubernetes, Git, CI/CD, Postman, n8n, Agile

EXPERIENCE

Cloud Engineering Intern (Backend Systems)

Aug 2025 – Jan 2026

IPAGE Group

Remote

- Architected 30+ serverless microservices on AWS Lambda powering an 8-step company registration and AI template generation workflow for dronetv.in, designed for 5,000 concurrent users serving 500–1,000 daily registrations.
- Reduced AI content generation costs by **88%** through intelligent model routing across GPT-3.5 and GPT-4 based on request complexity, triggered inline with form submission Lambda to eliminate separate execution cycles.
- Engineered a 3-stage DynamoDB pipeline (draft → under_review → approved) with S3-backed asset storage and state-driven Nodemailer triggers dispatching transactional emails at each workflow state transition.
- Built admin management backend with JWT-based auth and RBAC, enabling template approval, rejection, and live publishing directly to the DronTV company dashboard.

PROJECTS

MarketMind AI | Pinecone, Groq LLM, FinBERT, FastAPI, React | [GitHub](#)

Feb 2026 – Mar 2026

- Built a RAG-powered stock intelligence platform over 5 live sources (NSE, ET Markets, SEBI, BSE, YouTube) using Pinecone with BAAI/bge-base-en-v1.5 embeddings and MMR retrieval for diverse, non-redundant context.
- Implemented smart query routing stock-specific queries through Pinecone retrieval, general queries direct to Groq LLaMA-3.3-70b with FinBERT for real-time sentiment classification of market signals.
- Built a Finfluencer Fact-Checker that extracts financial claims from YouTube transcripts chunk-by-chunk, verifies each against live market data, and returns structured verdicts (Verified / Misleading / False / Unverifiable) with a video-level risk score.

Sahayak AI | AWS Lambda, Bedrock, DynamoDB, S3, EventBridge, OpenSearch, Python, React | [GitHub](#)

Sep 2025 – Jan 2026

- Built a multilingual AI classroom assistant for Indian middle schools (Grades 6–8) using 20+ AWS Lambda functions and Amazon Bedrock (Nova, Titan, Deepseek), supporting autonomous lesson scheduling, worksheet generation, assignment evaluation, and AI-powered doubt resolution.
- Engineered an NCERT data lake on S3 with OpenSearch Serverless vector indexing and RAG pipeline, reducing query response latency by **70%** through semantic retrieval over structured educational content.
- Implemented EventBridge-driven daily content pipeline delivering teacher-scheduled lessons enriched with Bedrock-generated analytical questions and YouTube recommendations per subject.

SafeX 3.0 | AWS Rekognition, Lambda, API Gateway, S3, MongoDB, Python, React | [GitHub](#)

Feb 2025 – Apr 2025

- Designed a 3-factor authentication system (bcrypt password + AWS Rekognition facial recognition at **99% accuracy** + OTP) orchestrated via 8 REST API endpoints on API Gateway with face images in S3 and metadata in MongoDB.
- Implemented serverless auth flow on AWS Lambda with IAM role-based access, signed S3 URLs for secure vault access, and CloudWatch logging across all functions.

CERTIFICATIONS & ACHIEVEMENTS

- **AWS Certified Solutions Architect – Associate** — Jun 2026 (Score: 893/1000)
- **AWS Certified Cloud Practitioner** — Apr 2026 (Score: 1000/1000)
- **IBM DevOps & Software Engineering Professional Certificate** — Feb 2026
- **Top Performing Candidate** — ET AI Hackathon 2026, The Economic Times (May 2026)
- **Winner** — Nebula Nexus Hackathon, MIT Jaipur (Jul 2025)